BUSINESS RESEARCH METHODS

Complete Exam Preparation Notes

PGDM – Business Analytics | Exam-oriented revision

Exam pattern: 50 marks, 2 hours. Section A: 10×1=10; Section B: answer any 5 of Q11–Q17 (5×3=15); Section C: answer any 3 cases (3×5=15); Section D: answer any 2 (2×5=10).

Important: These notes include the topics discussed from the supplied BRM reference material, plus clearly labeled exam-oriented explanations for the concepts requested in the conversation.

1. INTRODUCTION TO BUSINESS RESEARCH

Research: Research is an organized and systematic way of finding answers to questions.

Business Research: An organized, systematic and objective process of gathering, recording, analyzing and interpreting data to provide information for business decision-making.

Main objective: To provide accurate and timely information to management so that effective decisions can be made.

Basic Research: Knowledge/theory oriented; increases the knowledge base.

Applied Research: Practical problem solving; conducted to solve a specific managerial/business problem.

Quantitative Research: Uses numerical data and statistical analysis; generally structured and objective.

Qualitative Research: Seeks in-depth understanding of meanings, experiences, attitudes and the reasons behind behavior; often uses flexible methods.

Descriptive Research: Describes characteristics of a population or phenomenon—who, what, when, where and how. It does not by itself establish cause and effect.

Exploratory Research: Used when the problem is unclear; provides initial insights, helps define the problem and develop hypotheses.

Causal Research: Studies cause-and-effect relationships.

2. RESEARCH PROCESS

Sequence to memorize: Problem selection → Literature review → Define/specify problem → Objectives/hypothesis → Research design → Questionnaire/scales → Pilot test → Data collection → Data processing/analysis → Interpretation/reporting → Management decision.

Research problem: A condition in which some aspect of organizational performance is less than desirable, or an opportunity requiring a management decision.

Research objective: A clear statement of what the research intends to find out or achieve.

Hypothesis: A testable statement/assumption about a population, difference, effect or relationship.

3. RESEARCH DESIGN

Meaning: The overall plan/blueprint for conducting the research and obtaining the required information.

Key idea: It connects the research problem and objectives with data collection, measurement and analysis.

Qualitative design: Flexible, exploratory, focused on meanings and experiences.

Quantitative design: Structured, numerical and suitable for statistical analysis.

4. PRIMARY AND SECONDARY DATA

Primary data: First-hand data collected by the researcher for the current study. Examples: questionnaire, interview, observation, survey.

Secondary data: Data already collected for another purpose. Examples: reports, government publications, research papers, company records and databases.

Memory: Primary = I collect it. Secondary = already collected.

5. OBSERVATION AND SURVEY

Observation: Researcher watches and records actual behavior.

Advantages of observation: Behavior is recorded directly and can be relatively objective; useful for studying actual actions.

Limitations: Cannot easily observe attitudes, motives or mental states; can be time-consuming and costly.

Survey: Collects information by asking people questions; commonly uses a formal questionnaire.

Survey modes in the notes: Telephone, personal, mail and electronic interviews.

Survey advantages: Broad range of information, relatively fast/cheap compared with observation, simple administration and analysis.

Survey limitations: Unwillingness, lack of knowledge or memory, inability to explain motives, respondent bias and semantic difficulties.

6. QUESTIONNAIRE DESIGN

Questionnaire: A formalized set of questions for obtaining information from respondents.

Design sequence: Specify information needed → overcome respondent inability/unwillingness → determine content → decide structure → wording → order → form/layout → interviewing method → pre-test.

Good questionnaire: Not too lengthy; simple language; clear, sensible and answerable questions.

Avoid: Ambiguous words, leading questions, unnecessary calculations/generalizations and illogical ordering.

Question order: Start with interesting/simple/non-threatening questions; general before specific; difficult or sensitive questions later; use careful branching.

Pre-test: Test content, wording, sequence, layout, difficulty and instructions with respondents similar to the actual sample. The notes mention an initial small pre-test of about 15–30 respondents.

7. QUESTIONNAIRE VS SCHEDULE

Questionnaire: The respondent/informant fills the form.

Schedule: The investigator/enumerator asks the questions and fills the form.

Cost: Questionnaire collection is relatively cheap/economical; schedule collection is relatively more expensive.

Non-response: Complex questionnaire questions may produce non-response; a schedule reduces this problem because an enumerator records the answers.

Identity: With a questionnaire it may not always be clear who replies; with a schedule the respondent's identity is known.

Memory: Questionnaire = respondent writes. Schedule = surveyor writes.

8. MEASUREMENT AND SCALING

Measurement: Assigning numbers or symbols to characteristics of objects according to pre-specified rules.

Scaling: Creating a continuum on which measured objects/respondents are located.

Continuum: A continuous range from one extreme to another. Example: 1 = extremely unfavorable and 100 = extremely favorable.

Difference: Measurement is the actual assignment of a number; scaling is the process of placing the respondent/object on the continuum.

9. FOUR PRIMARY SCALES — NOIR

Nominal: Classification/identification only. Example: brand number or gender category. Memory: Nominal = Name.

Ordinal: Shows rank/order but not the magnitude of differences. Example: 1st, 2nd, 3rd. Memory: Ordinal = Order.

Interval: Equal intervals, but no absolute zero. Example: Fahrenheit/Celsius temperature. Differences are meaningful; ratios are not.

Ratio: Has all lower-scale properties plus an absolute zero; ratios are meaningful. Examples: age, income, weight, sales, time.

NOIR: Nominal → Ordinal → Interval → Ratio.

10. SCALING TECHNIQUES

Comparative scales: Objects are directly compared; resulting data are generally ordinal/rank-order.

Paired Comparison: Two objects are presented at a time and the respondent chooses one. For n objects, comparisons = n(n–1)/2.

Rank Order: Several objects are ranked according to a criterion. For n objects, n–1 scaling decisions are needed.

Constant Sum: Respondent allocates a fixed total, commonly 100 points, among attributes according to importance.

Non-comparative scales: Each object is evaluated independently (monadic).

Continuous Rating: Respondent marks a position on a continuous line from one extreme to another.

Itemized Rating: Respondent selects a category with a number/description. Common types: Likert, Semantic Differential and Stapel.

Likert: Measures degree of agreement/disagreement, commonly 1–5.

Semantic Differential: Seven-point scale using bipolar/opposite adjectives such as Poor–Excellent or Unreliable–Reliable.

Stapel: Unipolar rating scale presented from –5 to +5, without a neutral zero category in the response scale.

11. DATA PREPARATION AND ANALYSIS

Data preparation: Includes checking/editing, coding, organizing and preparing data for analysis.

Descriptive statistics: Summarize data using mean, median, mode, range, variance and standard deviation.

Inferential statistics: Use sample information to draw conclusions; examples include hypothesis tests, confidence intervals and regression.

Interpretation: Making sense of the statistical findings and connecting them to the research problem/objectives.

12. HYPOTHESIS TESTING

H0: Null hypothesis: no significant difference/effect/relationship.

H1: Alternative hypothesis: significant difference/effect/relationship exists.

Alpha: Common significance levels are 0.05 and 0.01.

Decision rule: If $p < \alpha$, reject H0. If $p \geq \alpha$, fail to reject H0.

Steps: State H0/H1 $\rightarrow$ choose α $\rightarrow$ calculate test statistic $\rightarrow$ obtain p-value $\rightarrow$ compare p with α $\rightarrow$ decide $\rightarrow$ interpret in business/research terms.

13. PARAMETRIC TESTS

t-test: Used to compare means, especially between two groups in common applications.

Independent t-test example: Compare average satisfaction of customers using App A versus App B.

Paired t-test example: Compare the same employees' productivity before and after training.

ANOVA: Used to compare means across three or more groups.

ANOVA example: Compare average customer satisfaction in Hyderabad, Mumbai and Delhi.

14. NON-PARAMETRIC / RANK-BASED TESTS

Meaning: Rank-based/non-parametric tests use ranks/order and generally make fewer distributional assumptions than parametric tests.

Mann–Whitney U: Two independent groups. Example: satisfaction ranks for App A versus App B.

Wilcoxon signed-rank: Two related/paired measurements. Example: before versus after training for the same employees.

Kruskal–Wallis: Three or more independent groups. Example: satisfaction ranks in Hyderabad, Mumbai and Delhi.

Friedman: Three or more related groups. Example: the same customers rank Brand A, B and C.

Memory: 2 independent = Mann–Whitney; 2 related = Wilcoxon; 3+ independent = Kruskal–Wallis; 3+ related = Friedman.

15. CHI-SQUARE, CORRELATION AND REGRESSION

Chi-square: Useful for categorical data and testing association/independence. Example: gender and product preference.

Correlation: Measures the direction and strength of association between variables. Example: advertising expenditure and sales.

Regression: Models/predicts a dependent variable from one or more independent variables. Example: predicting sales from advertising expenditure.

Memory: Correlation = relationship. Regression = prediction.

16. RESEARCH REPORT WRITING

Preliminary section: Title, transmittal/authorization material, contents/lists and executive summary.

Main section: Problem definition, approach/research design, analysis, results, limitations, conclusions/recommendations and exhibits.

Good report: Reader-oriented, easy to follow, professional, objective, concise and supported with appropriate visuals.

17. RESEARCH ETHICS — EXAM REVISION

Core principles: Honesty, objectivity, informed participation/consent where applicable, confidentiality/privacy, avoiding fabrication or falsification, accurate reporting and proper acknowledgment of sources.

Avoid: Plagiarism, manipulating results, inventing data, misleading participants or misrepresenting findings.

SECTION B — READY-TO-WRITE 3-MARK ANSWERS

The supplied exam-pattern file gives the Section-B structure (Q11–Q17, answer any five), but it does not contain the actual wording of those questions. The answers below are high-probability, syllabus-based 3-mark answers.

1. Differentiate between Questionnaire and Schedule.

A questionnaire is a formal set of questions filled in by the respondent, whereas a schedule is filled in by the investigator/enumerator while asking questions. Questionnaire collection is relatively cheaper and may have more non-response; schedule collection is relatively more expensive but reduces non-response and identifies the respondent more clearly. Example: Google Form filled by a patient = questionnaire; interviewer records the patient's answers = schedule.

2. What is Scaling? Explain Continuum.

Scaling is the process of creating a continuum on which respondents or objects are located according to a characteristic. A continuum is a continuous range from one extreme to another. Example: customer attitude can be measured from 1 = extremely unfavorable to 100 = extremely favorable. Measurement assigns the number; scaling places the respondent on the continuum.

3. Explain the four primary scales of measurement.

Nominal classifies objects; ordinal gives rank/order; interval has equal intervals but no absolute zero; ratio has equal intervals and an absolute zero. Examples: brand category (nominal), rank position (ordinal), temperature (interval), income/age/weight (ratio). Memory: NOIR.

4. Explain Paired Comparison Scaling.

Paired comparison presents two objects at a time and asks the respondent to choose one according to a criterion. Example: Coke versus Pepsi—choose the preferred brand. For n objects, the number of comparisons is $n(n–1)/2$. The resulting data are ordinal.

5. Explain Rank Order Scaling.

Rank order scaling asks respondents to arrange several objects according to a criterion. Example: rank four smartphone brands from most preferred to least preferred. It provides relative order but not the magnitude of differences, so the data are ordinal.

6. What is Constant Sum Scaling?

In constant sum scaling, respondents distribute a fixed number of points, commonly 100, among attributes according to importance. Example: Price 30, camera 25, battery 20, performance 15 and design 10. The total must equal 100.

7. Explain Likert Scale.

A Likert scale measures the degree of agreement or disagreement with statements. A common format is 1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree and 5 = strongly agree. It is commonly used for attitudes and opinions.

8. Differentiate Likert, Semantic Differential and Stapel scales.

Likert measures agreement/disagreement; Semantic Differential uses bipolar opposite adjectives on a seven-point scale, such as Poor–Excellent; Stapel uses a numerical scale from –5 to +5. Thus, the three differ mainly in the way response categories are presented.

9. What are rank-based tests? Give examples.

Rank-based tests analyze observations using their ranks/order and are useful for ordinal data or when parametric assumptions are unsuitable. Examples are Mann–Whitney U for two independent groups, Wilcoxon signed-rank for two related groups, Kruskal–Wallis for 3+ independent groups and Friedman for 3+ related groups.

10. Differentiate Parametric and Non-parametric Tests.

Parametric tests generally use stronger assumptions about numerical data and distributions; examples include t-test and ANOVA. Non-parametric tests make fewer distributional assumptions and often work with ranks or categorical data; examples include Mann–Whitney, Wilcoxon, Kruskal–Wallis, Friedman and chi-square.

11. Explain t-test and ANOVA.

A t-test is commonly used to compare means between two groups or two related measurements. ANOVA is used to compare means across three or more groups. Example: App A versus App B = t-test; Hyderabad versus Mumbai versus Delhi = ANOVA.

12. Explain Chi-square Test.

Chi-square is used mainly with categorical data to examine association/independence between categorical variables. Example: a researcher can test whether gender is associated with product preference. The conclusion is based on the calculated test statistic/p-value.

13. Explain Primary and Secondary Data.

Primary data are collected first-hand by the researcher for the current study, such as interviews and questionnaires. Secondary data already exist because they were collected earlier for another purpose, such as government reports, company records and research papers.

14. Explain Observation and Survey.

Observation records actual behavior by watching respondents or events. Survey collects information by asking questions using a formal questionnaire. Observation is useful for actual behavior but cannot easily reveal motives; surveys can collect attitudes, opinions and motives but may suffer from response bias or non-response.

15. Explain Hypothesis Testing.

Hypothesis testing is a statistical procedure for making a decision about a population using sample evidence. H0 states no significant effect/difference, while H1 states a significant effect/difference. At $\alpha = 0.05$, if $p < 0.05$, reject H0; otherwise fail to reject H0.

RAPID REVISION — LAST-DAY SHEET

Concept	Remember
Research	Organized + systematic search for answers.
Business research	Data → information → business decision.
Process	Problem → Literature → Objectives/Hypothesis → Design → Instrument → Pilot → Data → Analysis → Ir
Questionnaire	Respondent fills.
Schedule	Enumerator fills.
Continuum	Range from one extreme to another.
Measurement	Assign numbers.
Scaling	Place on continuum.
NOIR	Nominal, Ordinal, Interval, Ratio.
Paired comparison	Two at a time; n(n–1)/2.
Rank order	Arrange objects; n–1 decisions.
Constant sum	Usually 100 points.
Likert	Agreement/disagreement.
Semantic differential	Bipolar adjectives; 7 points.
Stapel	–5 to +5.
t-test	2 means.
ANOVA	3+ means.
Chi-square	Categorical association.
Mann–Whitney	2 independent rank groups.
Wilcoxon	2 related rank groups.
Kruskal–Wallis	3+ independent rank groups.
Friedman	3+ related rank groups.
Correlation	Relationship.
Regression	Prediction.
p-value rule	$p < 0.05$ → reject H0.

Exam writing tip: For a 3-mark answer, write a definition, 2–3 key points, and one simple example. For a 5-mark answer/case, add application, interpretation and conclusion.